

RUDRA PATEL

☎ 224-607-9378 ✉ rudrap26176@gmail.com 💼 [linkedin.com/in/rudrapatel09](https://www.linkedin.com/in/rudrapatel09) 🌐 rudras-website.vercel.app

Education

Lewis University

May 2026

B.S. Computer Science - GPA 3.9, Dean's List 6x

Coursework: Data Structures & Algorithms, Database Systems (SQL, NoSQL), Object-Oriented Programming (Java)

Experience

Software Engineer

May 2025 - May 2026

Hometown Liquor - Local Retail Business

- Built and shipped a full stack e-commerce platform from scratch with **Java** (Spring Boot), **TypeScript** (React), supporting 150+ users and 770+ products while helping grow monthly revenue to \$25K
- Developed and secured 92+ backend REST API endpoints for authentication, orders, inventory, analytics, and coupons using JWT authentication, role-based access control, rate limiting, and request validation
- Implemented a product search system that runs **PostgreSQL** keyword search rankings and AI matching in parallel, delivering relevant catalog results to users in 500-600ms on average
- Built a daily data ingestion pipeline from a third party API by asynchronously processing 100 product batches, handling duplicate records and updating stock without overwriting existing store data
- Improved release reliability by containerizing backend services with **Docker**, adding **CI/CD** pipelines, and writing 85+ unit and integration tests covering authentication, inventory synchronization, and core API workflows

Undergraduate Research Assistant

April 2026 – June 2026

Lewis University - Security Science Lab

- Built an internal dataset management and search platform with **Go** and **TypeScript** (React) for 25+ lab researchers, centralizing dataset upload, download, documentation, search, and editing
- Improved dataset search results by 30-40% by integrating an internal LLM API into the ingestion pipeline to generate searchable summaries, labels, file statistics, and annotation metadata
- Indexed 100+ research datasets in **PostgreSQL**, enabling researchers to use structured filters and relevance ranking to discover reusable datasets without relying on exact file names or manual documentation
- Engineered large file handling in **Go** supporting uploads and downloads up to 50 GB, eliminating timeout failures that previously blocked research operations

Projects

Real Time Event Processor | *TypeScript, Express.js, Redis, TimescaleDB, Docker*

[Link](#)

- Built a real time event processing pipeline using **TypeScript** (Node.js), and **Redis** that ingested, processed, stored, and visualized 2.4M+ events over 24 hours with zero data loss, containerized with **Docker** for consistent deployment
- Designed a batch consumer on Redis Streams that grouped up to 100 events per database write, cutting database transactions by 60–70% and ensuring no data was lost even if the service crashed mid batch
- Developed an Express.js **REST API** for querying millions of event records and reduced dashboard latency 55x, from 1.8 seconds to 33ms, by precomputing results instead of repeatedly scanning raw data

AI Recipe Service | *Java, Spring Boot, AWS, PostgreSQL, Docker*

[Link](#)

- Built a **Java** (Spring Boot) backend with 12 REST API endpoints and a pluggable LLM integration layer supporting OpenAI, Anthropic, and local models, with token usage and request latency tracked for every response
- Added caching, rate limiting, retry handling, and failure recovery to improve reliability under model and provider failures, supported by 139 unit and integration tests
- Defined secure **AWS** infrastructure in Terraform with ECS Fargate, RDS PostgreSQL, IAM, S3, and CloudWatch to isolate compute, protect data, and monitor service health

Cross Platform CI Runner | *Go, Python, GitHub Actions, YAML*

[Link](#)

- Shipped an open-source CI runner in **Go** that lets developers reproduce GitHub Actions style workflows, validating YAML pipelines and running jobs with dependencies from the terminal on Windows, macOS, and Linux
- Implemented parallel job execution, running independent jobs like linting, unit tests, and integration tests at the same time while respecting dependencies, cutting benchmark runtime by 47.4%
- Made failures easier to debug by saving logs and artifacts for every job, generating HTML reports, adding reruns for failed jobs only, and grouping repeated failures to flag flaky or persistent problems

Technical Skills

Languages & Tools: Java, Go, Python, TypeScript, JavaScript, SQL, CI/CD, Linux, Git

Frameworks & APIs: Spring Boot, Node.js, Express.js, React, REST APIs, LLM APIs, Claude Code

Data & Infrastructure: PostgreSQL, Redis, MySQL, MongoDB, AWS, Docker, Kubernetes